

# TIC 326225355

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

## BLS EPHEMERIS

Period 15.291303 d (ALIASED, see below)  
-> corrected period: 30.582605 d  
Epoch (t0) 1632.8660 BTJD  
Duration 0.2440 d  
Depth 7.762 % (77,617 ppm)  
SNR 1242.8

## CATALOG CROSS-MATCH (novelty)

|                              |          |
|------------------------------|----------|
| VSX                          | no match |
| ASAS-SN                      | no match |
| Gaia eclipsing-binary table  | no match |
| Gaia variable classification | no match |
| TESS EB catalog (Prsa+2022)  | no match |
| ExoFOP TOI list              | no match |

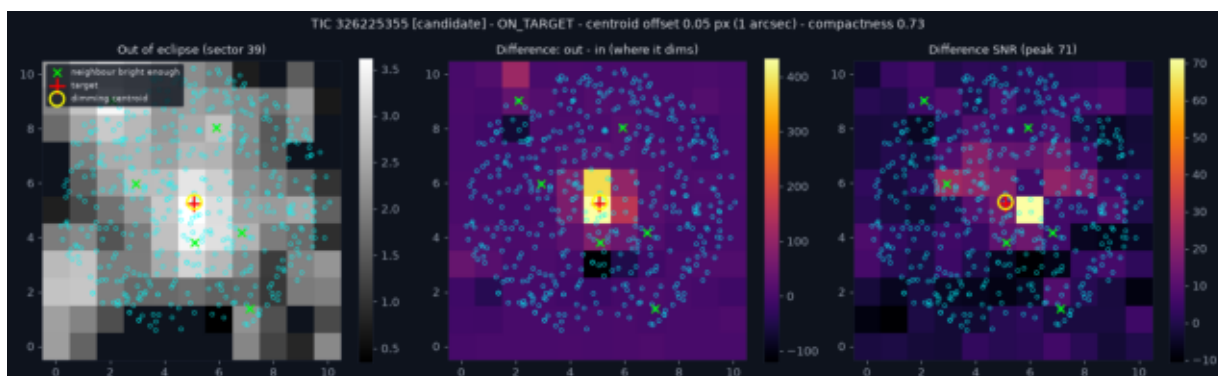
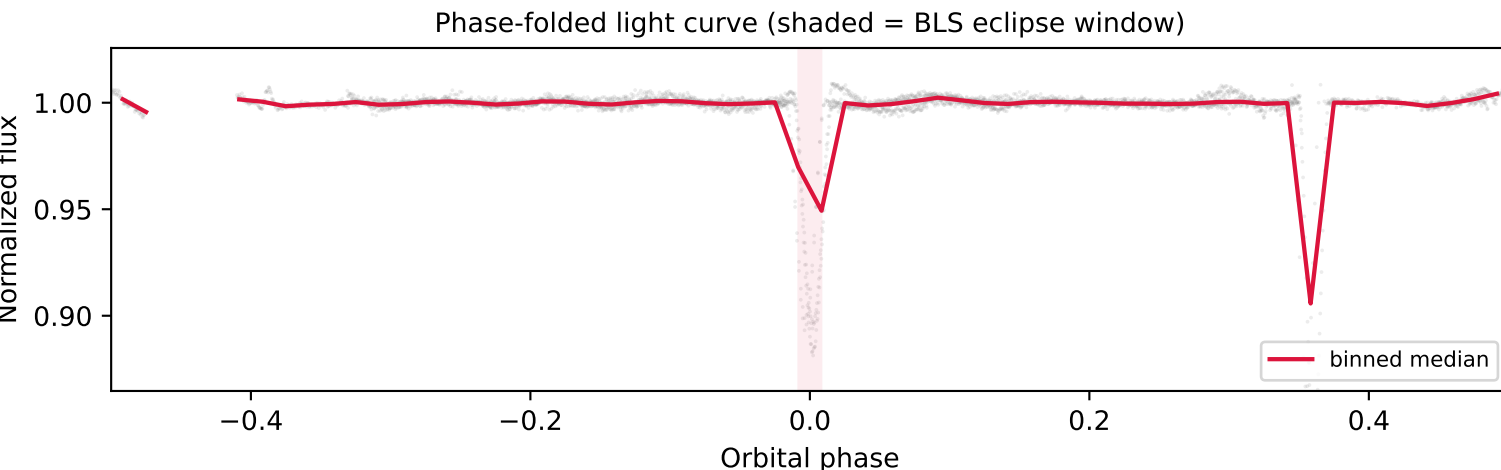
## INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate  $\geq 3$ )  
Robust depth ratio 1.12 (gate  $\geq 0.50$ )  
Robust depth sigma 424.9 (gate  $\geq 7.0$ )  
Odd/even z-score 3.56 (ALIAS FLAGGED)  
Secondary eclipse detected, phase 0.36, 6.8 sigma

Verdict: NOVEL

## PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 39 light-curve dip in this sector: 285.8 sigma t0 refined by 0.3600 phase  
Difference-image peak SNR 71.43 (gate  $\geq 5.0$ ) Compactness 0.73 (gate  $\geq 0.50$ )  
Dimming centroid offset 0.05 px (1 arcsec) (gate  $\leq 1.0$  px, i.e. on the target)  
Nearest bright-enough neighbour in cutout: TIC 326225457  
1.50 px away, dTmag +0.82 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 3.56 (gate  $> 3.0$ ): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 30.582605 d, in any write-up.